

Kinetic Modeling of Cell Signaling

Module 1: Reaction networks as dynamical systems

Jim Faeder

MSCBIO 2025 · Introduction to Bioinformatics Programming in Python

Fall 2026

Resources

- ▶ **BioNetGen** software, documentation, and tutorials:
bionetgen.org
- ▶ **VS Code extension:** search the Extensions panel for *BioNetGen*
- ▶ **BNG Playground:** write and simulate models in your browser, no installation
ruleworld.github.io/bngplayground
- ▶ **bngsim:** fast ODE and stochastic simulation from Python
bngsim.readthedocs.io · `pip install bngsim bionetgen`
- ▶ **Course repository:** models, notebooks, slides and handouts
github.com/jrfaeder/RBM_Intro_BioNetGen



Course repository



BNG Playground

Goals for this module

By the end of this module you should be able to

1. describe a biochemical process as a set of **species** and **reactions**

Goals for this module

By the end of this module you should be able to

1. describe a biochemical process as a set of **species** and **reactions**
2. write the rate of each reaction using **mass-action kinetics**

Goals for this module

By the end of this module you should be able to

1. describe a biochemical process as a set of **species** and **reactions**
2. write the rate of each reaction using **mass-action kinetics**
3. turn a reaction network into a system of **ordinary differential equations**

Goals for this module

By the end of this module you should be able to

1. describe a biochemical process as a set of **species** and **reactions**
2. write the rate of each reaction using **mass-action kinetics**
3. turn a reaction network into a system of **ordinary differential equations**
4. encode the same network in the **BioNetGen language** (BNGL), simulate it, and interpret the results

Goals for this module

By the end of this module you should be able to

1. describe a biochemical process as a set of **species** and **reactions**
2. write the rate of each reaction using **mass-action kinetics**
3. turn a reaction network into a system of **ordinary differential equations**
4. encode the same network in the **BioNetGen language** (BNGL), simulate it, and interpret the results
5. explain how a **negative feedback** loop shapes the dynamics of a signaling response

Part 1

Why model reaction networks?

Reaction networks describe how molecule populations change

A **reaction network model** consists of

- ▶ **species**: the molecules or complexes we track

Reaction networks describe how molecule populations change

A **reaction network model** consists of

- ▶ **species:** the molecules or complexes we track
- ▶ **reactions:** how species are converted into one another

Reaction networks describe how molecule populations change

A **reaction network model** consists of

- ▶ **species:** the molecules or complexes we track
- ▶ **reactions:** how species are converted into one another
- ▶ **rate laws and rate constants:** how fast each reaction occurs

Reaction networks describe how molecule populations change

A **reaction network model** consists of

- ▶ **species:** the molecules or complexes we track
- ▶ **reactions:** how species are converted into one another
- ▶ **rate laws** and **rate constants:** how fast each reaction occurs
- ▶ **initial conditions:** the amount of each species at $t = 0$

Reaction networks describe how molecule populations change

A **reaction network model** consists of

- ▶ **species**: the molecules or complexes we track
- ▶ **reactions**: how species are converted into one another
- ▶ **rate laws** and **rate constants**: how fast each reaction occurs
- ▶ **initial conditions**: the amount of each species at $t = 0$

The same framework describes atmospheric chemistry, epidemics, ecology, and the signaling and gene-regulatory networks that control cell behavior.

Reaction networks describe how molecule populations change

A **reaction network model** consists of

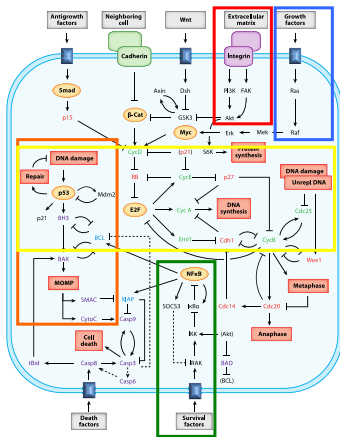
- ▶ **species:** the molecules or complexes we track
- ▶ **reactions:** how species are converted into one another
- ▶ **rate laws** and **rate constants:** how fast each reaction occurs
- ▶ **initial conditions:** the amount of each species at $t = 0$

The same framework describes atmospheric chemistry, epidemics, ecology, and the signaling and gene-regulatory networks that control cell behavior.

What a model gives us

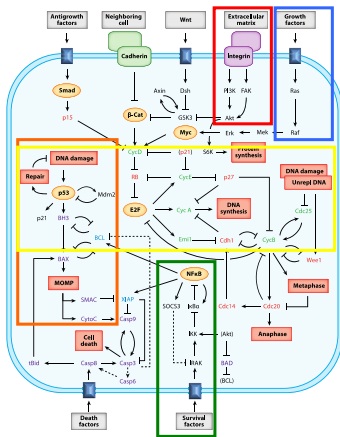
- ▶ A precise statement of a mechanism
- ▶ Predictions of dynamics we cannot easily measure
- ▶ A way to test whether a mechanism can explain data
- ▶ Hypotheses for the next experiment

Toward an integrated model of the cell



Cell fate is decided by many interacting signaling modules: growth-factor signaling, the cell cycle, DNA damage, apoptosis.

Toward an integrated model of the cell

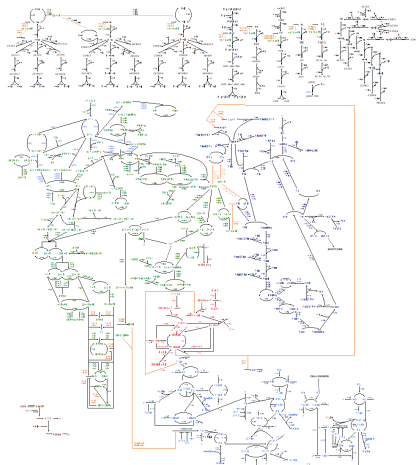
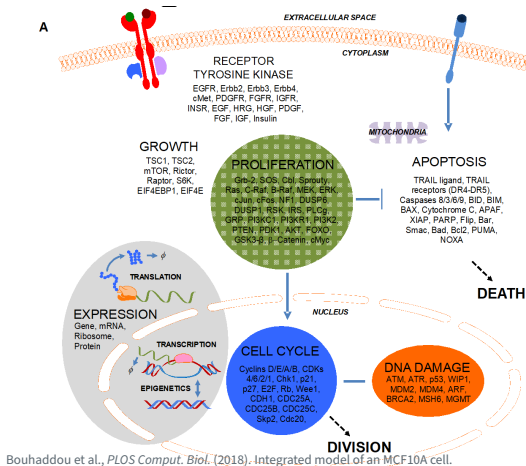


Cell fate is decided by many interacting signaling modules: growth-factor signaling, the cell cycle, DNA damage, apoptosis.

How do we link these modules together into models that predict how cells respond?

Tyson and Novak, *Annu. Rev. Phys. Chem.* **61**, 219–240 (2010)

Large models are built from simple reaction steps

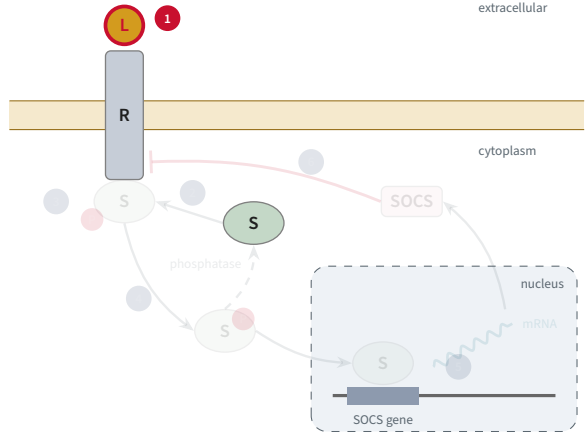


Part 2

A cytokine signaling circuit

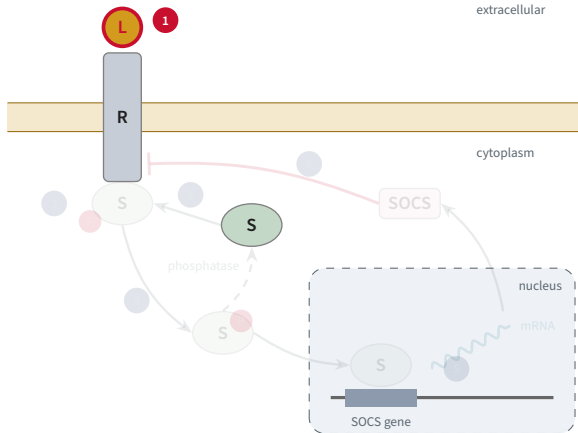
Cytokines signal through receptors, STATs, and SOCS feedback

- **Cytokines** (e.g. interleukins, interferons) are secreted proteins that coordinate immune responses.



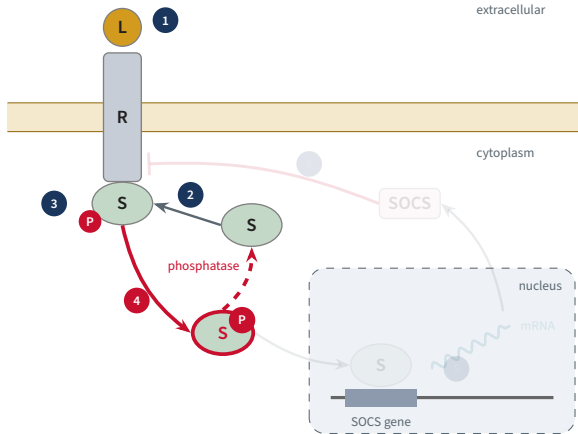
Cytokines signal through receptors, STATs, and SOCS feedback

- ▶ **Cytokines** (e.g. interleukins, interferons) are secreted proteins that coordinate immune responses.
- ▶ Binding to their **receptor** activates receptor-associated JAK kinases.



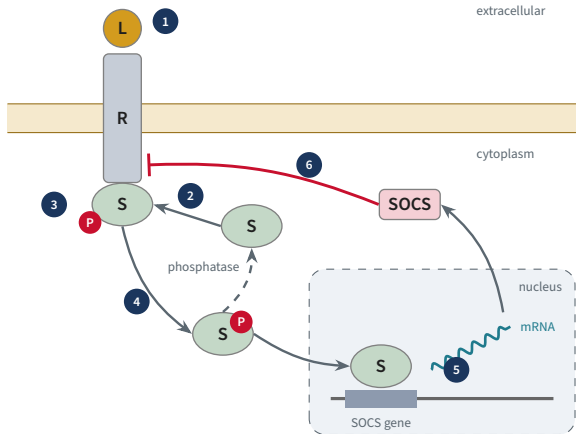
Cytokines signal through receptors, STATs, and SOCS feedback

- ▶ **Cytokines** (e.g. interleukins, interferons) are secreted proteins that coordinate immune responses.
- ▶ Binding to their **receptor** activates receptor-associated JAK kinases.
- ▶ **STAT** transcription factors dock on the active receptor, are phosphorylated, and move to the nucleus.



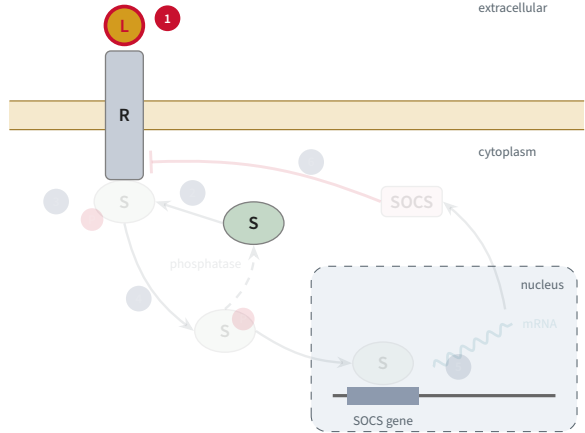
Cytokines signal through receptors, STATs, and SOCS feedback

- ▶ **Cytokines** (e.g. interleukins, interferons) are secreted proteins that coordinate immune responses.
- ▶ Binding to their **receptor** activates receptor-associated JAK kinases.
- ▶ **STAT** transcription factors dock on the active receptor, are phosphorylated, and move to the nucleus.
- ▶ pSTAT induces genes including **SOCS** (suppressor of cytokine signaling).
- ▶ SOCS binds the receptor and **shuts off further STAT activation**: negative feedback.



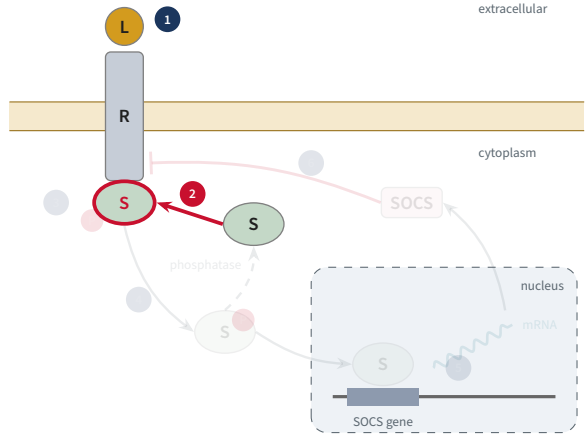
We will build the model one reaction step at a time

1. Ligand-receptor binding



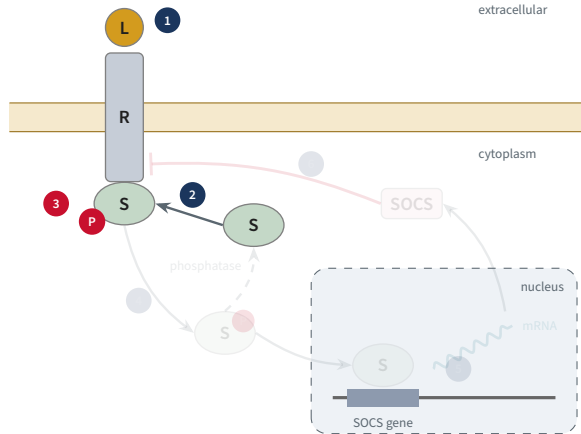
We will build the model one reaction step at a time

1. Ligand-receptor binding
2. STAT binds the ligand-bound receptor



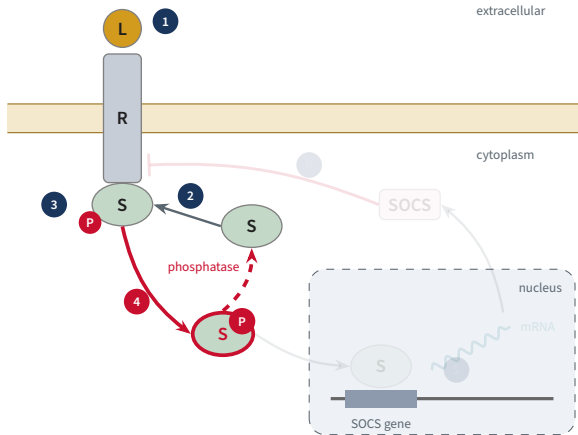
We will build the model one reaction step at a time

1. Ligand-receptor binding
2. STAT binds the ligand-bound receptor
3. Receptor-bound STAT is activated



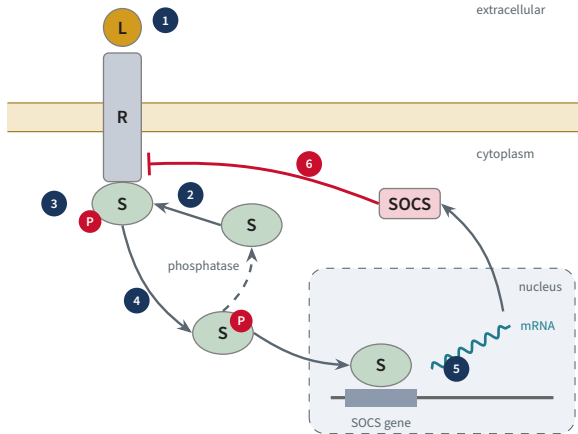
We will build the model one reaction step at a time

1. Ligand-receptor binding
2. STAT binds the ligand-bound receptor
3. Receptor-bound STAT is activated
4. Activated STAT dissociates



We will build the model one reaction step at a time

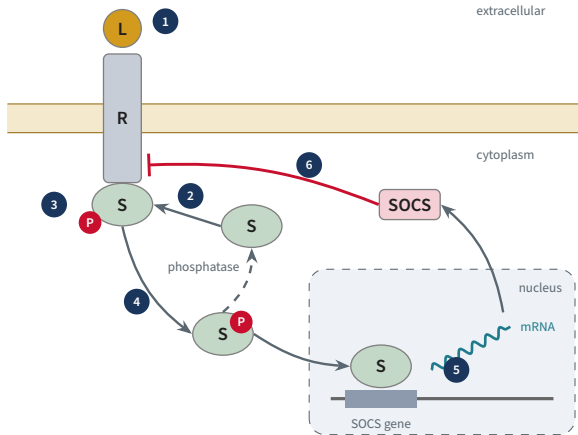
1. Ligand-receptor binding
2. STAT binds the ligand-bound receptor
3. Receptor-bound STAT is activated
4. Activated STAT dissociates
5. pSTAT binds the SOCS promoter; SOCS is transcribed and translated
6. SOCS binds the receptor and blocks STAT binding



We will build the model one reaction step at a time

1. Ligand-receptor binding
2. STAT binds the ligand-bound receptor
3. Receptor-bound STAT is activated
4. Activated STAT dissociates
5. pSTAT binds the SOCS promoter; SOCS is transcribed and translated
6. SOCS binds the receptor and blocks STAT binding

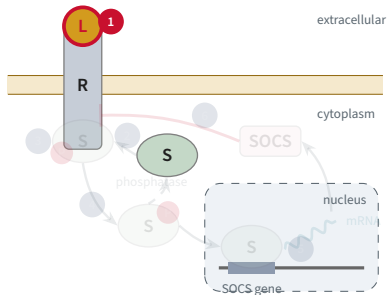
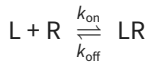
Each step adds reactions to the previous model.
The files are `models/stat_step1_binding.bngl`
through `stat_step6_feedback.bngl`.



Part 3

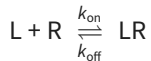
Step 1: Ligand–receptor binding

Binding is a reversible pair of reactions



Units: k_{on} in $\mu\text{M}^{-1}\text{min}^{-1}$, k_{off} in min^{-1}

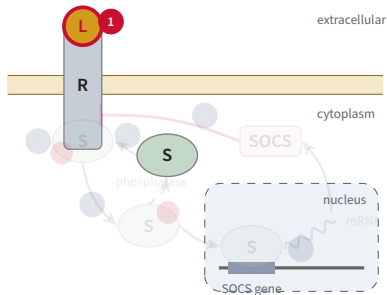
Binding is a reversible pair of reactions



Two elementary reactions, each with its own rate:

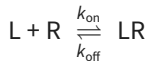
binding: $v_1 = k_{\text{on}} [L] [R]$

unbinding: $v_2 = k_{\text{off}} [LR]$



Units: k_{on} in $\mu\text{M}^{-1}\text{min}^{-1}$, k_{off} in min^{-1}

Binding is a reversible pair of reactions



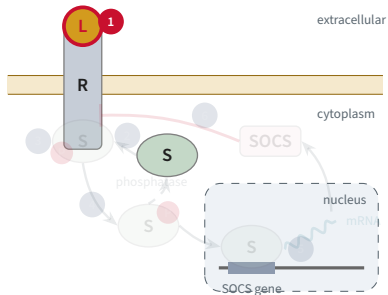
Two elementary reactions, each with its own rate:

binding: $v_1 = k_{on} [L] [R]$

unbinding: $v_2 = k_{\text{off}} [\text{LR}]$

Law of mass action

The rate of an elementary reaction is proportional to the product of the concentrations of its reactants. The proportionality constant is the **rate constant**.



Units: k_{on} in $\mu\text{M}^{-1}\text{min}^{-1}$, k_{off} in min^{-1}

Each reaction contributes to the rate of change of its species

	v_1 (bind)	v_2 (unbind)
L	-1	+1
R	-1	+1
LR	+1	-1

Reactants are consumed ($-$), products are created ($+$).

Each reaction contributes to the rate of change of its species

	v_1 (bind)	v_2 (unbind)
L	-1	+1
R	-1	+1
LR	+1	-1

Reactants are consumed (−), products are created (+).

$$\frac{d[L]}{dt} = -k_{\text{on}}[L][R] + k_{\text{off}}[LR]$$

$$\frac{d[R]}{dt} = -k_{\text{on}}[L][R] + k_{\text{off}}[LR]$$

$$\frac{d[LR]}{dt} = +k_{\text{on}}[L][R] - k_{\text{off}}[LR]$$

Each reaction contributes to the rate of change of its species

	v_1 (bind)	v_2 (unbind)
L	-1	+1
R	-1	+1
LR	+1	-1

Reactants are consumed (-), products are created (+).

$$\frac{d[L]}{dt} = -k_{\text{on}}[L][R] + k_{\text{off}}[LR]$$

$$\frac{d[R]}{dt} = -k_{\text{on}}[L][R] + k_{\text{off}}[LR]$$

$$\frac{d[LR]}{dt} = +k_{\text{on}}[L][R] - k_{\text{off}}[LR]$$

Ordinary differential equations (ODEs) follow mechanically from the list of reactions and their rate laws.

The stoichiometry matrix gives a compact general form

Collect the concentrations in a vector $\mathbf{X}$ and the reaction rates in $\mathbf{v}$:

$$\frac{d\mathbf{X}}{dt} = \mathbf{S} \mathbf{v}(\mathbf{X}), \quad \frac{d}{dt} \begin{pmatrix} [L] \\ [R] \\ [LR] \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & +1 \\ -1 & +1 \\ +1 & -1 \end{pmatrix}}_{\mathbf{S}} \underbrace{\begin{pmatrix} k_{\text{on}}[L][R] \\ k_{\text{off}}[LR] \end{pmatrix}}_{\mathbf{v}(\mathbf{X})}$$

The stoichiometry matrix gives a compact general form

Collect the concentrations in a vector $\mathbf{X}$ and the reaction rates in $\mathbf{v}$:

$$\frac{d\mathbf{X}}{dt} = \mathbf{S} \mathbf{v}(\mathbf{X}), \quad \frac{d}{dt} \begin{pmatrix} [L] \\ [R] \\ [LR] \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & +1 \\ -1 & +1 \\ +1 & -1 \end{pmatrix}}_{\mathbf{S}} \underbrace{\begin{pmatrix} k_{\text{on}}[L][R] \\ k_{\text{off}}[LR] \end{pmatrix}}_{\mathbf{v}(\mathbf{X})}$$

$$S_{ij} = (\text{times species } i \text{ is a product of reaction } j) - (\text{times species } i \text{ is a reactant of reaction } j)$$

The stoichiometry matrix gives a compact general form

Collect the concentrations in a vector $\mathbf{X}$ and the reaction rates in $\mathbf{v}$:

$$\frac{d\mathbf{X}}{dt} = \mathbf{S} \mathbf{v}(\mathbf{X}), \quad \frac{d}{dt} \begin{pmatrix} [L] \\ [R] \\ [LR] \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & +1 \\ -1 & +1 \\ +1 & -1 \end{pmatrix}}_{\mathbf{S}} \underbrace{\begin{pmatrix} k_{\text{on}}[L][R] \\ k_{\text{off}}[LR] \end{pmatrix}}_{\mathbf{v}(\mathbf{X})}$$

S_{ij} = (times species i is a product of reaction j) – (times species i is a reactant of reaction j)

- $\mathbf{S}$ has one row per species and one column per reaction.

The stoichiometry matrix gives a compact general form

Collect the concentrations in a vector $\mathbf{X}$ and the reaction rates in $\mathbf{v}$:

$$\frac{d\mathbf{X}}{dt} = \mathbf{S} \mathbf{v}(\mathbf{X}), \quad \frac{d}{dt} \begin{pmatrix} [L] \\ [R] \\ [LR] \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & +1 \\ -1 & +1 \\ +1 & -1 \end{pmatrix}}_{\mathbf{S}} \underbrace{\begin{pmatrix} k_{\text{on}}[L][R] \\ k_{\text{off}}[LR] \end{pmatrix}}_{\mathbf{v}(\mathbf{X})}$$

S_{ij} = (times species i is a product of reaction j) – (times species i is a reactant of reaction j)

- ▶ $\mathbf{S}$ has one row per species and one column per reaction.
- ▶ This form holds for *any* reaction network; only $\mathbf{S}$ and $\mathbf{v}$ change.

Conservation laws and equilibrium

Conservation. Binding only moves molecules between free and bound forms:

$$[L] + [LR] = L_0$$

$$[R] + [LR] = R_0$$

so only one of the three ODEs is independent.

These appear as linear dependencies among the rows of **S**.

Conservation laws and equilibrium

Conservation. Binding only moves molecules between free and bound forms:

$$[L] + [LR] = L_0$$

$$[R] + [LR] = R_0$$

so only one of the three ODEs is independent.

These appear as linear dependencies among the rows of **S**.

Equilibrium. At steady state $v_1 = v_2$:

$$\frac{[L][R]}{[LR]} = \frac{k_{\text{off}}}{k_{\text{on}}} \equiv K_D$$

Conservation laws and equilibrium

Conservation. Binding only moves molecules between free and bound forms:

$$[L] + [LR] = L_0$$

$$[R] + [LR] = R_0$$

so only one of the three ODEs is independent.

These appear as linear dependencies among the rows of **S**.

Equilibrium. At steady state $v_1 = v_2$:

$$\frac{[L][R]}{[LR]} = \frac{k_{\text{off}}}{k_{\text{on}}} \equiv K_D$$

Substituting the conservation laws gives a quadratic for $[LR]$:

$$[LR]_{\text{eq}} = \frac{1}{2} \left[b - \sqrt{b^2 - 4L_0R_0} \right],$$

$$b = L_0 + R_0 + K_D$$

Conservation laws and equilibrium

Conservation. Binding only moves molecules between free and bound forms:

$$[L] + [LR] = L_0$$

$$[R] + [LR] = R_0$$

so only one of the three ODEs is independent.

These appear as linear dependencies among the rows of $\mathbf{S}$.

Equilibrium. At steady state $v_1 = v_2$:

$$\frac{[L][R]}{[LR]} = \frac{k_{\text{off}}}{k_{\text{on}}} \equiv K_D$$

Substituting the conservation laws gives a quadratic for $[LR]$:

$$[LR]_{\text{eq}} = \frac{1}{2} \left[b - \sqrt{b^2 - 4L_0R_0} \right],$$

$$b = L_0 + R_0 + K_D$$

Small models can be solved by hand. Real signaling networks cannot, so we solve the ODEs numerically.

ODEs for reaction networks are solved numerically

Simplest idea (forward Euler): take small time steps

Δt ,

$$\mathbf{X}(t + \Delta t) \approx \mathbf{X}(t) + \Delta t \mathbf{S} \mathbf{v}(\mathbf{X}(t))$$

ODEs for reaction networks are solved numerically

Simplest idea (forward Euler): take small time steps Δt ,

$$\mathbf{X}(t + \Delta t) \approx \mathbf{X}(t) + \Delta t \mathbf{S} \mathbf{v}(\mathbf{X}(t))$$

In practice:

- The equations are **nonlinear** and usually have no closed-form solution.

ODEs for reaction networks are solved numerically

Simplest idea (forward Euler): take small time steps Δt ,

$$\mathbf{X}(t + \Delta t) \approx \mathbf{X}(t) + \Delta t \mathbf{S} \mathbf{v}(\mathbf{X}(t))$$

In practice:

- ▶ The equations are **nonlinear** and usually have no closed-form solution.
- ▶ Reactions occur on very different time scales, which makes the system **stiff**: simple methods need impractically small steps.

ODEs for reaction networks are solved numerically

Simplest idea (forward Euler): take small time steps Δt ,

$$\mathbf{X}(t + \Delta t) \approx \mathbf{X}(t) + \Delta t \mathbf{S} \mathbf{v}(\mathbf{X}(t))$$

In practice:

- ▶ The equations are **nonlinear** and usually have no closed-form solution.
- ▶ Reactions occur on very different time scales, which makes the system **stiff**: simple methods need impractically small steps.
- ▶ Production solvers (CVODE, LSODA) adapt the step size and use implicit methods for stiff systems.

ODEs for reaction networks are solved numerically

Simplest idea (forward Euler): take small time steps Δt ,

$$\mathbf{X}(t + \Delta t) \approx \mathbf{X}(t) + \Delta t \mathbf{S} \mathbf{v}(\mathbf{X}(t))$$

In practice:

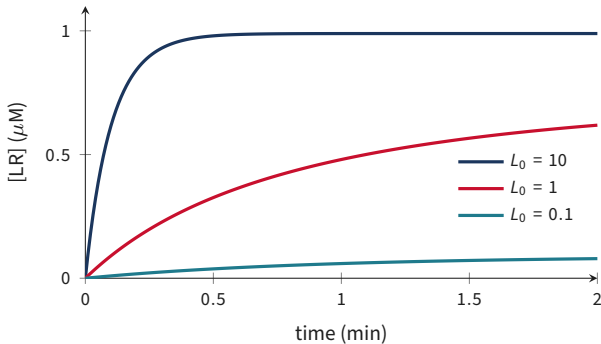
- ▶ The equations are **nonlinear** and usually have no closed-form solution.
- ▶ Reactions occur on very different time scales, which makes the system **stiff**: simple methods need impractically small steps.
- ▶ Production solvers (CVODE, LSODA) adapt the step size and use implicit methods for stiff systems.

In BioNetGen

```
simulate({method=>"ode", ...})  
uses CVODE.
```

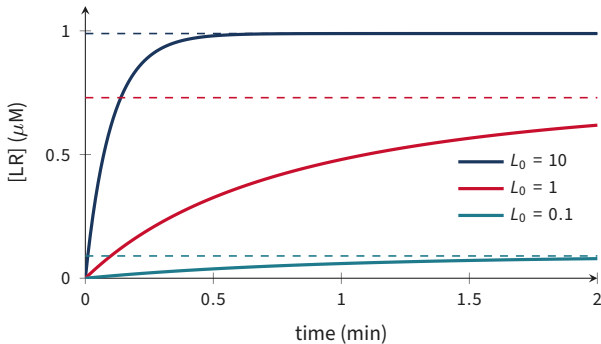
In Python, `bngsim.Simulator(model, method="ode")` runs the same model with CVODE.

Binding approaches equilibrium faster and further at higher ligand



$$R_0 = 1 \mu\text{M}, k_{\text{on}} = 1 \mu\text{M}^{-1}\text{min}^{-1}, \\ k_{\text{off}} = 0.1 \text{ min}^{-1} (K_D = 0.1 \mu\text{M}).$$

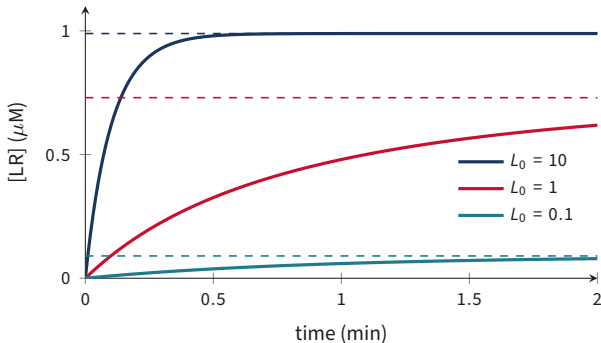
Binding approaches equilibrium faster and further at higher ligand



$$R_0 = 1 \mu\text{M}, k_{\text{on}} = 1 \mu\text{M}^{-1}\text{min}^{-1}, \\ k_{\text{off}} = 0.1 \text{ min}^{-1} (K_D = 0.1 \mu\text{M}).$$

Dashed lines: $[LR]_{\text{eq}}$ from the formula on the previous slide.

Binding approaches equilibrium faster and further at higher ligand



$$R_0 = 1 \mu\text{M}, k_{\text{on}} = 1 \mu\text{M}^{-1}\text{min}^{-1}, \\ k_{\text{off}} = 0.1 \text{ min}^{-1} (K_D = 0.1 \mu\text{M}).$$

Dashed lines: $[LR]_{\text{eq}}$ from the formula on the previous slide.

The relaxation rate is $k_{\text{on}}([L] + [R]) + k_{\text{off}}$ near equilibrium, so higher ligand also means a faster response.

The same model in BNGL

```
begin model

begin parameters
  L0      10      # ligand (uM)
  R0      1       # receptor (uM)
  kon     1       # binding (1/uM/min)
  koff    0.1     # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

The same model in BNGL

```
begin model

begin parameters
  L0      10    # ligand (uM)
  R0      1     # receptor (uM)
  kon     1     # binding (1/uM/min)
  koff    0.1   # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

```
begin observables
  Molecules Lfree  L
  Molecules Rfree  R
  Molecules LR     LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <=> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
```

The same model in BNGL

```
begin model

begin parameters
  L0      10      # ligand (uM)
  R0      1       # receptor (uM)
  kon     1       # binding (1/uM/min)
  koff    0.1     # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

```
begin observables
  Molecules Lfree  L
  Molecules Rfree  R
  Molecules LR     LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <=> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
```

► **observables:** quantities to output

The same model in BNGL

```
begin model

begin parameters
  L0      10      # ligand (uM)
  R0      1       # receptor (uM)
  kon     1       # binding (1/uM/min)
  koff    0.1     # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L      L0
  R      R0
end species
```

```
begin observables
  Molecules Lfree  L
  Molecules Rfree  R
  Molecules LR     LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <-> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
```

- ▶ **observables:** quantities to output
- ▶ **reaction rules:** $\leftrightarrow$ is reversible; forward and reverse rate constants

The same model in BNGL

```
begin model

begin parameters
  L0      10      # ligand (uM)
  R0      1       # receptor (uM)
  kon     1       # binding (1/uM/min)
  koff    0.1     # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

```
begin observables
  Molecules Lfree  L
  Molecules Rfree  R
  Molecules LR     LR
end observables

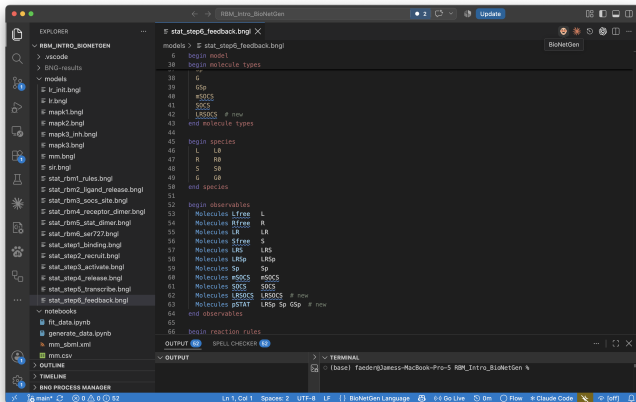
begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <-> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode", t_end=>2, n_steps=>200})
```

- ▶ **observables:** quantities to output
- ▶ **reaction rules:** $\leftrightarrow$ is reversible; forward and reverse rate constants
- ▶ **actions** after `end model`: generate the network, then simulate

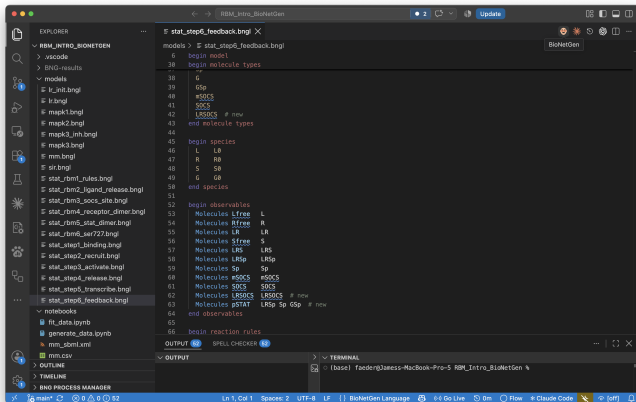
Running a model in VS Code



► Open a .bngl file

VS Code extension: Ali Sinan Saglam

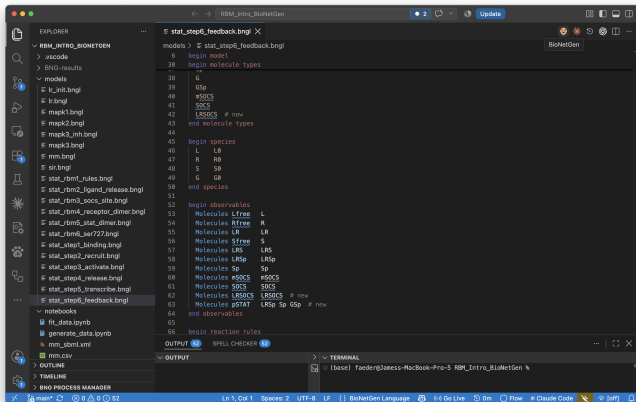
Running a model in VS Code



- ▶ Open a `.bngl` file
- ▶ Click the **run** button in the editor toolbar

VS Code extension: Ali Sinan Saglam

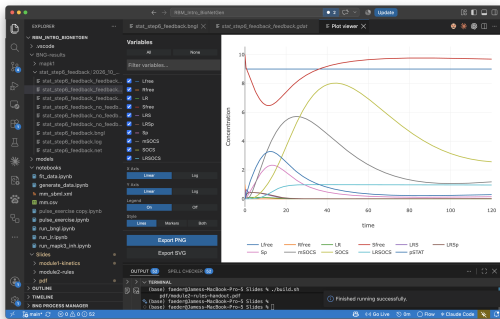
Running a model in VS Code



- ▶ Open a .bngl file
- ▶ Click the **run** button in the editor toolbar
- ▶ Results go to BNG-results/<model>/<timestamp>/

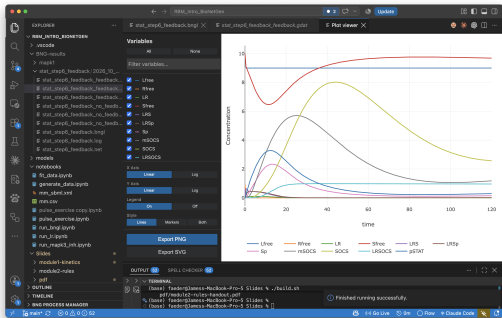
VS Code extension: Ali Sinan Saglam

Interactive plotting and Python notebooks

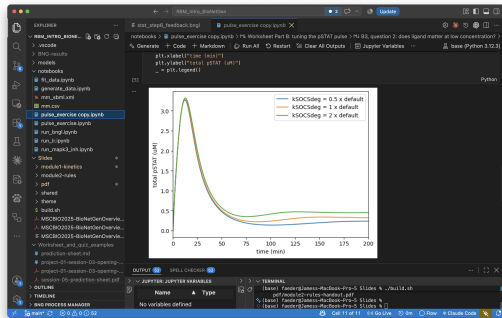


Built-in plotting of .gdat output

Interactive plotting and Python notebooks



Built-in plotting of .gdat output



Deeper analysis in Jupyter with bngsim

BNG Playground: BioNetGen in your browser

The screenshot displays the BNG Playground web application. On the left is the **BNGL Editor** with a text area containing a BNGL model for a binding reaction. The model includes parameters for initial molecule counts, association/dissociation rate constants, and a simulation command. On the right is the **Regulatory Graph** view, which visualizes the model as a network. It shows two reactant nodes, **B(a)** and **A(b)**, both pointing to a central reaction node **_R1**, which in turn points to the product node **AB(a+b)**. The interface includes tabs for Time Courses, Network, Analysis, and Regulatory, as well as a legend at the bottom right.

```
1 begin model
2   begin parameters
3     AB 100 # Initial number of A molecules
4     BB 100 # Initial number of B molecules
5     ka 0.1 # A-B association rate constant (1/molecule 1/s)
6     kd 1 # A-B dissociation rate constant (1/s)
7   end parameters
8   begin molecule types
9     A(b)
10    B(a)
11  end molecule types
12  begin seed species
13    A(b) AB
14    B(a) BB
15  end seed species
16  begin observables
17    Molecules A A(b)
18    Molecules B B(a)
19    Molecules C A(b)11,B(a)11
20  end observables
21  begin reaction rules
22    A(b) + B(a) <=> A(b)11,B(a)11 ka, kd
23  end reaction rules
24 end model
25
26 simulate(method="cde",t_end=10,n_steps=200)
27
```

BNGL editor (left) and the regulatory graph of a binding rule (right)

- No installation: write, simulate, and analyze models



ruleworld.github.io/bngplayground

BNG Playground: BioNetGen in your browser

The screenshot displays the BNG Playground web interface. On the left is the BNGL Editor, which contains a text area with BNGL code for a binding model. The code defines parameters for initial molecule counts, association/dissociation rates, molecule types, seed species, observables, and reaction rules. Below the code area are buttons for 'Models', 'Load', 'Export', 'Format', 'Parse', 'Run', and 'Auto'. On the right is the Regulatory Graph, titled 'About Regulatory Graph'. It shows a network diagram where two input nodes, 'B(a)' and 'A(b)', both point to a central node 'J1'. This node 'J1' then points to an output node 'AB(1),B(a)1'. The interface includes a top navigation bar with tabs for 'Time Courses', 'Network', 'Analysis', 'Regulatory', 'Contact Map', 'Rules', 'Influence', and 'Analysis'. A 'Share' button is also present in the top right.

```
1 begin model
2   begin parameters
3     AB 100 # Initial number of A molecules
4     DB 100 # Initial number of B molecules
5     ka 0.1 # A-B association rate constant (1/molecule 1/s)
6     kd 1 # A-B dissociation rate constant (1/s)
7   end parameters
8   begin molecule types
9     A(b)
10    B(a)
11  end molecule types
12  begin seed species
13    A(b) AB
14    B(a) DB
15  end seed species
16  begin observables
17    Molecules A A(b)
18    Molecules B B(a)
19    Molecules C A(b)1, B(a)1
20  end observables
21  begin reaction rules
22    A(b) + B(a) <=> A(b)1, B(a)1 ka, kd
23  end reaction rules
24 end model
25
26 simulate(method="cde", t_end=10, n_steps=200)
27
```

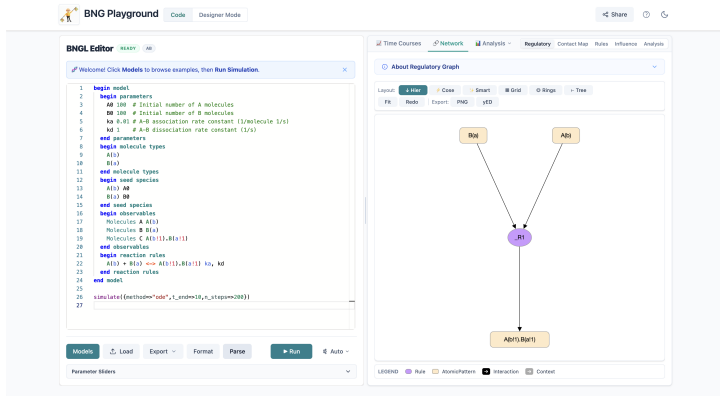
BNGL editor (left) and the regulatory graph of a binding rule (right)

- ▶ No installation: write, simulate, and analyze models
- ▶ Built-in examples



ruleworld.github.io/bngplayground

BNG Playground: BioNetGen in your browser



BNGL editor (left) and the regulatory graph of a binding rule (right)

- ▶ No installation: write, simulate, and analyze models
- ▶ Built-in examples
- ▶ Share a model as a link



ruleworld.github.io/bngplayground

BNG Playground: BioNetGen in your browser

The screenshot displays the BNG Playground web interface. On the left is the BNGL Editor, which contains a text area with a BNGL model definition. The model defines parameters for initial numbers of molecules (A0, B0), association and dissociation rate constants (ka, kd), and a reaction rule for the binding of A and B to form a complex AB. Below the text area are buttons for 'Models', 'Load', 'Export', 'Format', 'Parse', 'Run', and 'Auto'. On the right is the Regulatory Graph view, titled 'About Regulatory Graph'. It shows a network diagram where two nodes, 'B(a)' and 'A(b)', are connected by arrows to a central node 'J1'. This node 'J1' is then connected by an arrow to a final node 'AB(1),B(a)1'. The interface also includes a 'Share' button at the top right and a 'Legend' at the bottom right of the graph area.

BNGL editor (left) and the regulatory graph of a binding rule (right)

- ▶ No installation: write, simulate, and analyze models
- ▶ Built-in examples
- ▶ Share a model as a link
- ▶ Paste in any file from models/

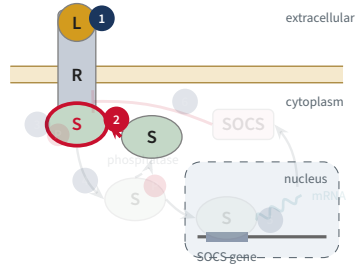
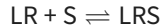


ruleworld.github.io/bngplayground

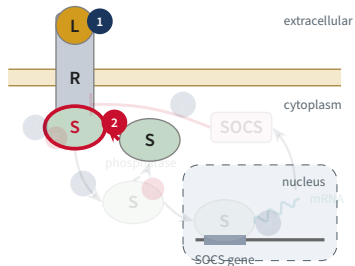
Part 4

Building the signaling circuit in BNGL

Step 2: STAT binds the ligand-bound receptor



Step 2: STAT binds the ligand-bound receptor



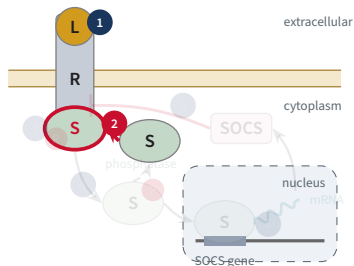
STAT can bind *only* the ligand-bound receptor LR, never free R. This is how ligand binding **enables** STAT recruitment.

Step 2: STAT binds the ligand-bound receptor



New lines in `stat_step2_recruit.bngl`:

```
begin parameters
  S0      10    # STAT (uM)
  kSon    1     # STAT binding (1/uM/min)
  kSoff   1     # STAT unbinding (1/min)
end parameters
begin molecule types
  S
  LRS
end molecule types
begin species
  S S0
end species
begin reaction rules
  Sbind: LR + S <-> LRS kSon, kSoff
end reaction rules
```



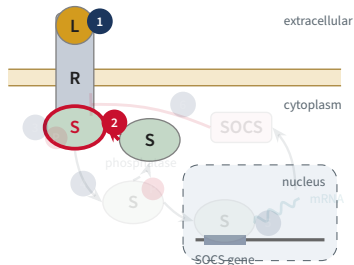
STAT can bind *only* the ligand-bound receptor LR, never free R. This is how ligand binding **enables** STAT recruitment.

Step 2: STAT binds the ligand-bound receptor



New lines in `stat_step2_recruit.bngl`:

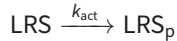
```
begin parameters
  S0      10    # STAT (uM)
  kSon    1     # STAT binding (1/uM/min)
  kSoff   1     # STAT unbinding (1/min)
end parameters
begin molecule types
  S
  LRS
end molecule types
begin species
  S S0
end species
begin reaction rules
  Sbind: LR + S <-> LRS kSon, kSoff
end reaction rules
```



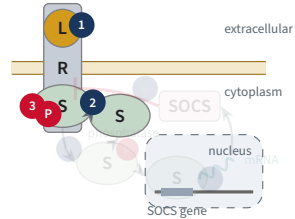
STAT can bind *only* the ligand-bound receptor LR, never free R. This is how ligand binding **enables** STAT recruitment.

Every new species needs a molecule type; species present at $t = 0$ also need an initial amount.

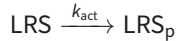
Step 3: receptor-bound STAT is activated



A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\text{act}}[\text{LRS}]$.



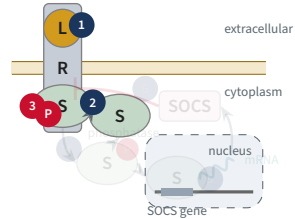
Step 3: receptor-bound STAT is activated



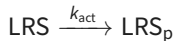
A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\text{act}}[\text{LRS}]$.

```
kact 1 # STAT activation (1/min)
LRSp
Sact: LRS -> LRSp kact
```

one new line each in parameters, molecule types, and reaction rules



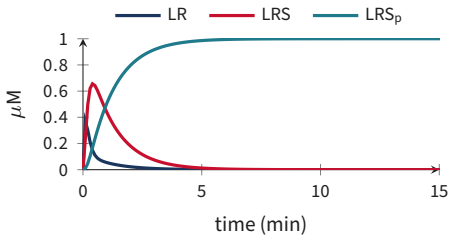
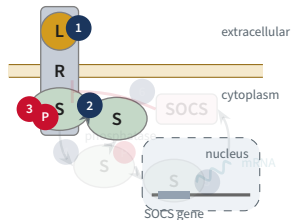
Step 3: receptor-bound STAT is activated



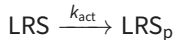
A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\text{act}}[\text{LRS}]$.

```
kact 1 # STAT activation (1/min)
LRSp
Sact: LRS -> LRSp kact
```

one new line each in parameters, molecule types, and reaction rules



Step 3: receptor-bound STAT is activated

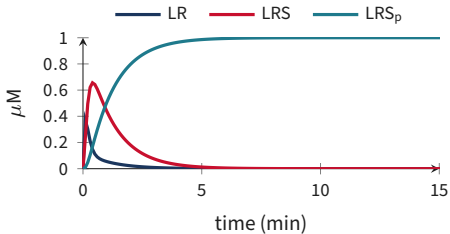
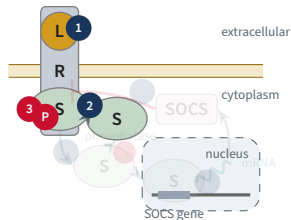


A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\text{act}}[\text{LRS}]$.

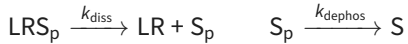
```
kact 1 # STAT activation (1/min)
LRSp
Sact: LRS -> LRSp kact
```

one new line each in parameters, molecule types, and reaction rules

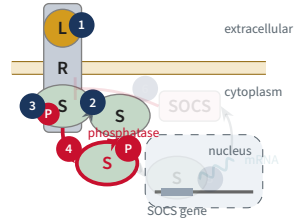
Problem: pSTAT never leaves the receptor. Every receptor ends up stuck as LRS_p.



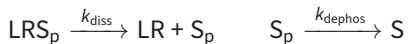
Step 4: activated STAT dissociates from the receptor



Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

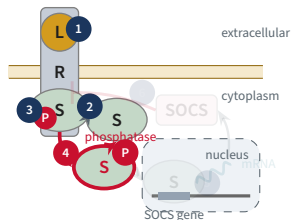


Step 4: activated STAT dissociates from the receptor

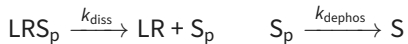


Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

```
kdiss    1    # pSTAT release (1/min)
kdephos  0.1  # dephosphorylation (1/min)
Sp
Srelease: LRSp -> LR + Sp kdiss
Sdephos:  Sp -> S kdephos
```

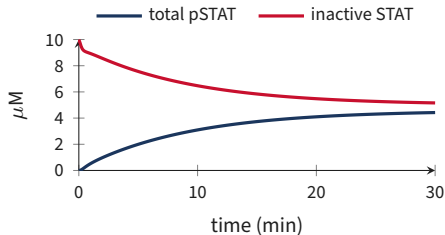
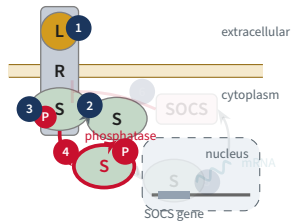


Step 4: activated STAT dissociates from the receptor

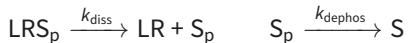


Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

```
kdiss    1    # pSTAT release (1/min)
kdephos  0.1  # dephosphorylation (1/min)
Sp
Srelease: LRSp -> LR + Sp kdiss
Sdephos:  Sp -> S kdephos
```



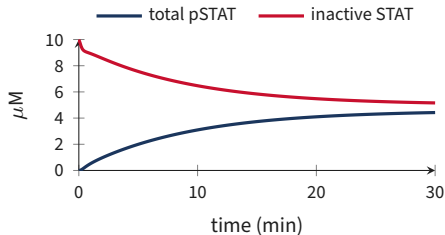
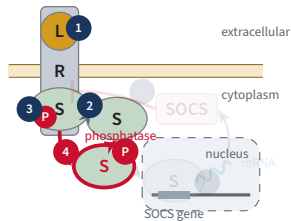
Step 4: activated STAT dissociates from the receptor



Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

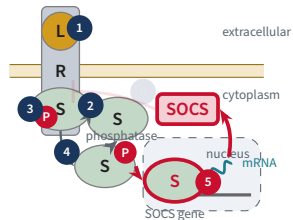
```
kdiss    1    # pSTAT release (1/min)
kdephos  0.1  # dephosphorylation (1/min)
Sp
Srelease: LRSp -> LR + Sp kdiss
Sdephos:  Sp -> S kdephos
```

pSTAT now reaches a steady state set by the balance of activation and dephosphorylation.



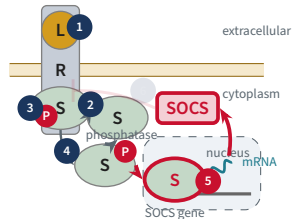
Step 5: pSTAT induces SOCS expression

```
# pSTAT binds the SOCS promoter
Gbind: Sp + G <=> GSp kGon, kGoff
# Bound promoter is transcribed
Transcribe: GSp -> GSp + mSOCS ktx
# mRNA degradation
mRNAdeg: mSOCS -> 0 kmdeg
# Translation
Translate: mSOCS -> mSOCS + SOCS kt1
# SOCS protein degradation
SOCSdeg: SOCS -> 0 kSOCSdeg
```



Step 5: pSTAT induces SOCS expression

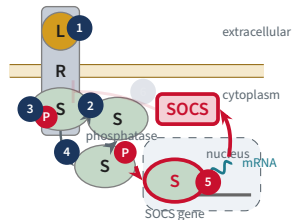
```
# pSTAT binds the SOCS promoter
Gbind: Sp + G <-> GSp kGon, kGoff
# Bound promoter is transcribed
Transcribe: GSp -> GSp + mSOCS ktx
# mRNA degradation
mRNAdeg: mSOCS -> 0 kmdeg
# Translation
Translate: mSOCS -> mSOCS + SOCS kt1
# SOCS protein degradation
SOCSdeg: SOCS -> 0 kSOCSdeg
```



- **Synthesis:** a species on both sides (GSp, mSOCS) is a template and is not consumed.

Step 5: pSTAT induces SOCS expression

```
# pSTAT binds the SOCS promoter
Gbind: Sp + G <-> GSp kGon, kGoff
# Bound promoter is transcribed
Transcribe: GSp -> GSp + mSOCS ktx
# mRNA degradation
mRNAdeg: mSOCS -> 0 kmdeg
# Translation
Translate: mSOCS -> mSOCS + SOCS kt1
# SOCS protein degradation
SOCSdeg: SOCS -> 0 kSOCSdeg
```

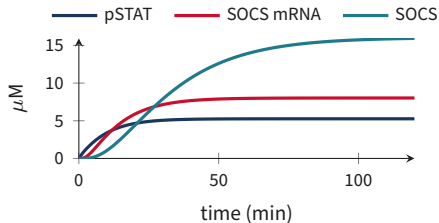
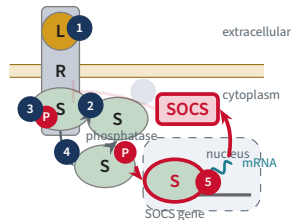


- **Synthesis:** a species on both sides (GSp, mSOCS) is a template and is not consumed.
- **Degradation:** 0 means nothing; the molecule is removed.

Step 5: pSTAT induces SOCS expression

```
# pSTAT binds the SOCS promoter
Gbind: Sp + G <-> GSp kGon, kGoff
# Bound promoter is transcribed
Transcribe: GSp -> GSp + mSOCS ktx
# mRNA degradation
mRNAdeg: mSOCS -> 0 kmdeg
# Translation
Translate: mSOCS -> mSOCS + SOCS kt1
# SOCS protein degradation
SOCSdeg: SOCS -> 0 kSOCSdeg
```

- **Synthesis:** a species on both sides (GSp, mSOCS) is a template and is not consumed.
- **Degradation:** 0 means nothing; the molecule is removed.

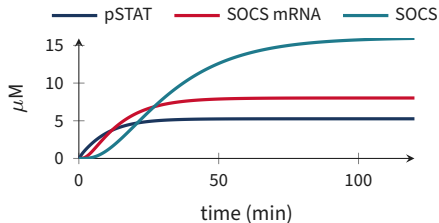
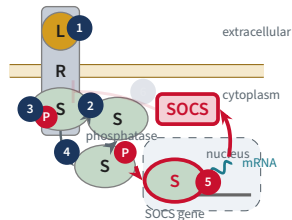


Step 5: pSTAT induces SOCS expression

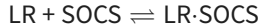
```
# pSTAT binds the SOCS promoter
Gbind: Sp + G <-> GSp kGon, kGoff
# Bound promoter is transcribed
Transcribe: GSp -> GSp + mSOCS ktx
# mRNA degradation
mRNAdeg: mSOCS -> 0 kmdeg
# Translation
Translate: mSOCS -> mSOCS + SOCS kt1
# SOCS protein degradation
SOCSdeg: SOCS -> 0 kSOCSdeg
```

- **Synthesis:** a species on both sides (GSp, mSOCS) is a template and is not consumed.
- **Degradation:** 0 means nothing; the molecule is removed.

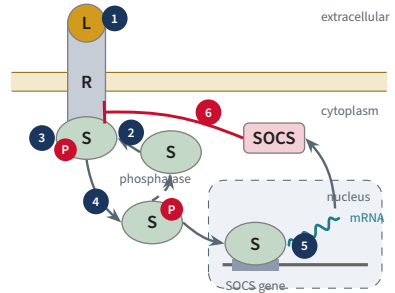
SOCS accumulates, but it has no target yet, so pSTAT is unaffected.



Step 6: SOCS binds the receptor and blocks STAT recruitment



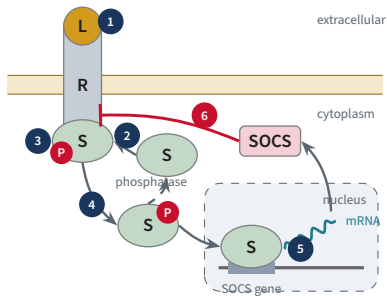
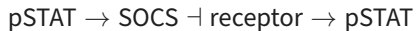
SOCS competes with STAT for the active receptor. LR·SOCS cannot recruit STAT, so pSTAT production drops.



Step 6: SOCS binds the receptor and blocks STAT recruitment



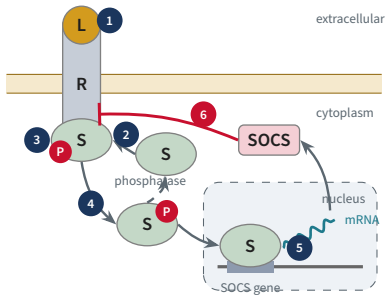
SOCS competes with STAT for the active receptor. LR·SOCS cannot recruit STAT, so pSTAT production drops. The loop is closed:



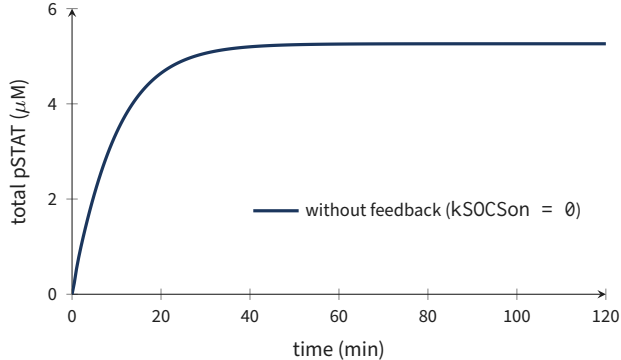
```

kSOCSon      1      # (1/uM/min)
kSOCSoFF     0.01   # (1/min)
LRSOCS
SOCSbind: LR + SOCS <=> LRSOCS kSOCSon, kSOCSoFF

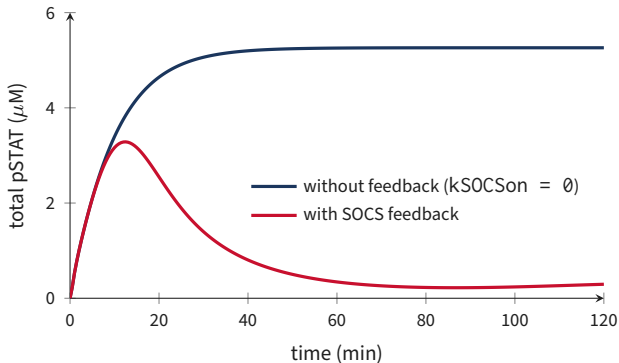
```



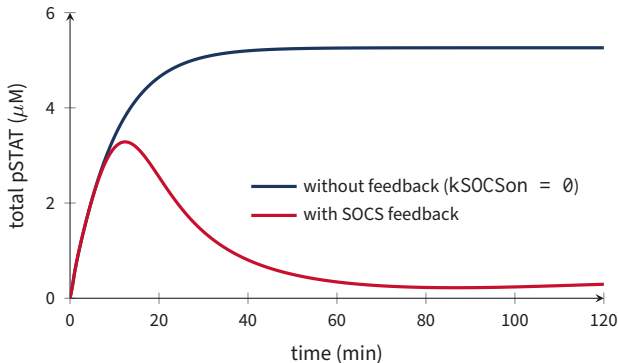
Negative feedback turns a sustained signal into a pulse



Negative feedback turns a sustained signal into a pulse

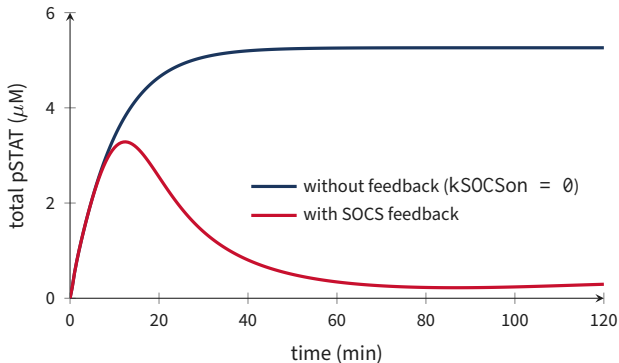


Negative feedback turns a sustained signal into a pulse



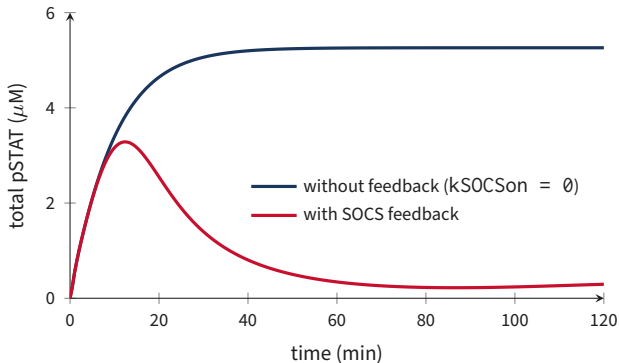
- Early response is identical: SOCS has not been made yet.

Negative feedback turns a sustained signal into a pulse



- ▶ Early response is identical: SOCS has not been made yet.
- ▶ As SOCS accumulates it sequesters receptors, and pSTAT falls to a low level despite continued ligand.

Negative feedback turns a sustained signal into a pulse



- ▶ Early response is identical: SOCS has not been made yet.
- ▶ As SOCS accumulates it sequesters receptors, and pSTAT falls to a low level despite continued ligand.
- ▶ The delay comes from transcription and translation.

Both curves come from `stat_step6_feedback.bng1`, which runs the model twice using `setParameter`.

The complete network

	reaction	rate
1	$L + R \rightleftharpoons LR$	k_{on}, k_{off}
2	$LR + S \rightleftharpoons LRS$	k_{Son}, k_{Soff}
3	$LRS \rightarrow LRS_p$	k_{act}
4	$LRS_p \rightarrow LR + S_p$	k_{diss}
	$S_p \rightarrow S$	k_{dephos}
5	$S_p + G \rightleftharpoons GS_p$	k_{Gon}, k_{Goff}
	$GS_p \rightarrow GS_p + mSOCS$	k_{tx}
	$mSOCS \rightarrow 0$	k_{mdeg}
	$mSOCS \rightarrow mSOCS + SOCS$	k_{tl}
	$SOCS \rightarrow 0$	$k_{SOCSdeg}$
6	$LR + SOCS \rightleftharpoons LR \cdot SOCS$	$k_{SOCSon}, k_{SOCSoff}$

The complete network

	reaction	rate
1	$L + R \rightleftharpoons LR$	k_{on}, k_{off}
2	$LR + S \rightleftharpoons LRS$	k_{Son}, k_{Soff}
3	$LRS \rightarrow LRS_p$	k_{act}
4	$LRS_p \rightarrow LR + S_p$	k_{diss}
	$S_p \rightarrow S$	k_{dephos}
5	$S_p + G \rightleftharpoons GS_p$	k_{Gon}, k_{Goff}
	$GS_p \rightarrow GS_p + mSOCS$	k_{tx}
	$mSOCS \rightarrow 0$	k_{mdeg}
	$mSOCS \rightarrow mSOCS + SOCS$	k_{tl}
	$SOCS \rightarrow 0$	$k_{SOCSdeg}$
6	$LR + SOCS \rightleftharpoons LR \cdot SOCS$	$k_{SOCSon}, k_{SOCSoff}$

- 11 lines of BNGL generate **15 reactions** (4 are reversible) among **12 species**.

The complete network

	reaction	rate
1	$L + R \rightleftharpoons LR$	k_{on}, k_{off}
2	$LR + S \rightleftharpoons LRS$	k_{Son}, k_{Soff}
3	$LRS \rightarrow LRS_p$	k_{act}
4	$LRS_p \rightarrow LR + S_p$	k_{diss}
	$S_p \rightarrow S$	k_{dephos}
5	$S_p + G \rightleftharpoons GS_p$	k_{Gon}, k_{Goff}
	$GS_p \rightarrow GS_p + mSOCS$	k_{tx}
	$mSOCS \rightarrow 0$	k_{mdeg}
	$mSOCS \rightarrow mSOCS + SOCS$	k_{tl}
	$SOCS \rightarrow 0$	$k_{SOCSdeg}$
6	$LR + SOCS \rightleftharpoons LR \cdot SOCS$	$k_{SOCSon}, k_{SOCSoff}$

- ▶ 11 lines of BNGL generate **15 reactions** (4 are reversible) among **12 species**.
- ▶ **S** is a 12×15 matrix; BioNetGen builds it and the ODEs for you (see the .net file).

The complete network

	reaction	rate
1	$L + R \rightleftharpoons LR$	k_{on}, k_{off}
2	$LR + S \rightleftharpoons LRS$	k_{Son}, k_{Soff}
3	$LRS \rightarrow LRS_p$	k_{act}
4	$LRS_p \rightarrow LR + S_p$	k_{diss}
	$S_p \rightarrow S$	k_{dephos}
5	$S_p + G \rightleftharpoons GS_p$	k_{Gon}, k_{Goff}
	$GS_p \rightarrow GS_p + mSOCS$	k_{tx}
	$mSOCS \rightarrow 0$	k_{mdeg}
	$mSOCS \rightarrow mSOCS + SOCS$	k_{tl}
	$SOCS \rightarrow 0$	$k_{SOCSdeg}$
6	$LR + SOCS \rightleftharpoons LR \cdot SOCS$	$k_{SOCSon}, k_{SOCSoff}$

- ▶ 11 lines of BNGL generate **15 reactions** (4 are reversible) among **12 species**.
- ▶ **S** is a 12×15 matrix; BioNetGen builds it and the ODEs for you (see the .net file).
- ▶ Every species had to be named by hand: L, LR, LRS, LRS_p, LR·SOCS, ...

Part 5

Looking ahead

Naming every complex by hand does not scale

Our model made strong simplifying assumptions. Relax a few of them:

- ▶ SOCS can also bind receptors that already carry STAT (LRS, LRS_p)

Naming every complex by hand does not scale

Our model made strong simplifying assumptions. Relax a few of them:

- ▶ SOCS can also bind receptors that already carry STAT (LRS, LRS_p)
- ▶ Ligand can dissociate from *any* receptor complex, not only LR

Naming every complex by hand does not scale

Our model made strong simplifying assumptions. Relax a few of them:

- ▶ SOCS can also bind receptors that already carry STAT (LRS, LRS_p)
- ▶ Ligand can dissociate from *any* receptor complex, not only LR
- ▶ The receptor has two chains, each recruiting its own STAT

Naming every complex by hand does not scale

Our model made strong simplifying assumptions. Relax a few of them:

- ▶ SOCS can also bind receptors that already carry STAT (LRS, LRS_p)
- ▶ Ligand can dissociate from *any* receptor complex, not only LR
- ▶ The receptor has two chains, each recruiting its own STAT
- ▶ STATs dimerize before entering the nucleus

Naming every complex by hand does not scale

Our model made strong simplifying assumptions. Relax a few of them:

- ▶ SOCS can also bind receptors that already carry STAT (LRS, LRS_p)
- ▶ Ligand can dissociate from *any* receptor complex, not only LR
- ▶ The receptor has two chains, each recruiting its own STAT
- ▶ STATs dimerize before entering the nucleus

Each feature **multiplies** the number of distinct complexes, and each complex needs its own species name and its own copy of every reaction it takes part in.

Naming every complex by hand does not scale

Our model made strong simplifying assumptions. Relax a few of them:

- ▶ SOCS can also bind receptors that already carry STAT (LRS, LRS_p)
- ▶ Ligand can dissociate from *any* receptor complex, not only LR
- ▶ The receptor has two chains, each recruiting its own STAT
- ▶ STATs dimerize before entering the nucleus

Each feature **multiplies** the number of distinct complexes, and each complex needs its own species name and its own copy of every reaction it takes part in.

Module 2: represent molecules with internal structure (binding sites and states) and write rules that generate the network automatically.

ODEs describe average behavior

- ▶ ODEs treat concentrations as continuous and deterministic.

ODEs describe average behavior

- ▶ ODEs treat concentrations as continuous and deterministic.
- ▶ In a single cell, some species are present in very small numbers: one or two copies of a gene, a few mRNA molecules.

ODEs describe average behavior

- ▶ ODEs treat concentrations as continuous and deterministic.
- ▶ In a single cell, some species are present in very small numbers: one or two copies of a gene, a few mRNA molecules.
- ▶ Then random timing of individual reactions matters, and cells with identical parameters respond differently.

ODEs describe average behavior

- ▶ ODEs treat concentrations as continuous and deterministic.
- ▶ In a single cell, some species are present in very small numbers: one or two copies of a gene, a few mRNA molecules.
- ▶ Then random timing of individual reactions matters, and cells with identical parameters respond differently.
- ▶ The **same reaction network** can be simulated stochastically with the Gillespie algorithm:
`simulate({method=>"ssa", ...}).`

ODEs describe average behavior

- ▶ ODEs treat concentrations as continuous and deterministic.
- ▶ In a single cell, some species are present in very small numbers: one or two copies of a gene, a few mRNA molecules.
- ▶ Then random timing of individual reactions matters, and cells with identical parameters respond differently.
- ▶ The **same reaction network** can be simulated stochastically with the Gillespie algorithm:
`simulate({method=>"ssa", ...}).`

We return to stochastic simulation in a later module.

End-of-class exercises

On the worksheet ([Worksheets/module1-exercises.pdf](#)):

- A. Check the ligand–receptor equilibrium against the formula.

End-of-class exercises

On the worksheet (`Worksheets/module1-exercises.pdf`):

- A. Check the ligand–receptor equilibrium against the formula.
- B. **Tune the pSTAT pulse:** predict, then test, how six parameters change the time of the peak, its height, and the adapted level.

End-of-class exercises

On the worksheet ([Worksheets/module1-exercises.pdf](#)):

- A. Check the ligand–receptor equilibrium against the formula.
- B. **Tune the pSTAT pulse:** predict, then test, how six parameters change the time of the peak, its height, and the adapted level.
- C. Remove the phosphatase. What sets the final level of pSTAT?

End-of-class exercises

On the worksheet ([Worksheets/module1-exercises.pdf](#)):

- A. Check the ligand–receptor equilibrium against the formula.
- B. **Tune the pSTAT pulse:** predict, then test, how six parameters change the time of the peak, its height, and the adapted level.
- C. Remove the phosphatase. What sets the final level of pSTAT?
- D. Let SOCS bind STAT-loaded receptors. What new species and reactions do you need?

End-of-class exercises

On the worksheet ([Worksheets/module1-exercises.pdf](#)):

- A. Check the ligand–receptor equilibrium against the formula.
- B. **Tune the pSTAT pulse:** predict, then test, how six parameters change the time of the peak, its height, and the adapted level.
- C. Remove the phosphatase. What sets the final level of pSTAT?
- D. Let SOCS bind STAT-loaded receptors. What new species and reactions do you need?

No local setup yet? Paste any model from [models/](#) into [BNG Playground](#) ([ruleworld.github.io/bngplayground](#)) and run it in your browser.

Summary

- ▶ A reaction network is defined by its species, reactions, rate laws, and initial conditions.

Summary

- ▶ A reaction network is defined by its species, reactions, rate laws, and initial conditions.
- ▶ Mass-action rate laws turn the network into ODEs: $d\mathbf{X}/dt = \mathbf{S} \mathbf{v}(\mathbf{X})$.

Summary

- ▶ A reaction network is defined by its species, reactions, rate laws, and initial conditions.
- ▶ Mass-action rate laws turn the network into ODEs: $d\mathbf{X}/dt = \mathbf{S} \mathbf{v}(\mathbf{X})$.
- ▶ BNGL describes the network directly; BioNetGen generates and integrates the ODEs.

Summary

- ▶ A reaction network is defined by its species, reactions, rate laws, and initial conditions.
- ▶ Mass-action rate laws turn the network into ODEs: $d\mathbf{X}/dt = \mathbf{S} \mathbf{v}(\mathbf{X})$.
- ▶ BNGL describes the network directly; BioNetGen generates and integrates the ODEs.
- ▶ Building the cytokine–STAT–SOCS circuit one step at a time shows how each mechanism contributes; negative feedback produces an adaptive pulse of pSTAT.

Summary

- ▶ A reaction network is defined by its species, reactions, rate laws, and initial conditions.
- ▶ Mass-action rate laws turn the network into ODEs: $d\mathbf{X}/dt = \mathbf{S} \mathbf{v}(\mathbf{X})$.
- ▶ BNGL describes the network directly; BioNetGen generates and integrates the ODEs.
- ▶ Building the cytokine–STAT–SOCS circuit one step at a time shows how each mechanism contributes; negative feedback produces an adaptive pulse of pSTAT.
- ▶ Writing every complex as a separate species quickly becomes unmanageable. This motivates rule-based modeling, the subject of Module 2.